



CPSC 100

Computational Thinking

Algorithm, Classifiers and Trees!

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Course Admin



Agenda

- Course Admin
- Recap of Pre-Reading
- Clicker Questions
- Classification
- Real world examples

Slides for Pre-reading

**Slides with yellow borders
will not be covered in
class, but is still testable
content - you should
review this before class.**

Transistors



A transistor is a building block of modern electronics. Tens of billions of transistors are jammed into a “chip” so that data can be stored and processed. [Read more.](#)

Moore's Law

Moore's Law

Gordon Moore one of the co-founders of **Intel** corporation established the term "Moore's Law" in 1965, This law explains how the number of transistors on integrated circuits is increasing exponentially, which boosts computing capability and lowers prices. Moore's law has had a significant and far-reaching impact on technology, from the introduction of personal computers and smartphones to the advancement of artificial intelligence and Internet of Things(IoT) devices. it has accelerated development in sectors including telecommunications, healthcare, transportation, enabling organizations and people to accomplish things that were previously unthinkable.

Moore's Law Definition

The exponential increase in the number of transistors on integrated circuits over time is referred to as Moore's law. According to this, a chip transistor count tends to double every two years or so, resulting in higher processing power and better performance.

Is Moore's Law still valid?

The Transistor Cliff

Sarah Constantin

Moore's law may be coming to an end. What happens to AI progress if it does?

How Hardware Affects AI Progress

The biggest AI models are trained on expensive, state-of-the-art microchips, or semiconductors. Only a few organizations, such as Google and OpenAI, have the budgets to train them. For years, improvements in AI performance have been driven by progress in this underlying hardware.

For most of the history of semiconductor manufacturing, steadily and predictably accelerating improvements in performance and reductions in price have been the norm. This pattern has been codified as "Moore's Law," Intel CEO Gordon Moore's observation that the number of transistors that could be placed on a chip for the same price doubled approximately every two years. That may be coming to an end. Depending on the specific semiconductor performance metric, Moore's Law has either stalled out already, or is on course to soon hit fundamental physical limits.

So, what could happen "after Moore's Law?" And how would that affect AI performance?

Let's zoom in and look at the details.

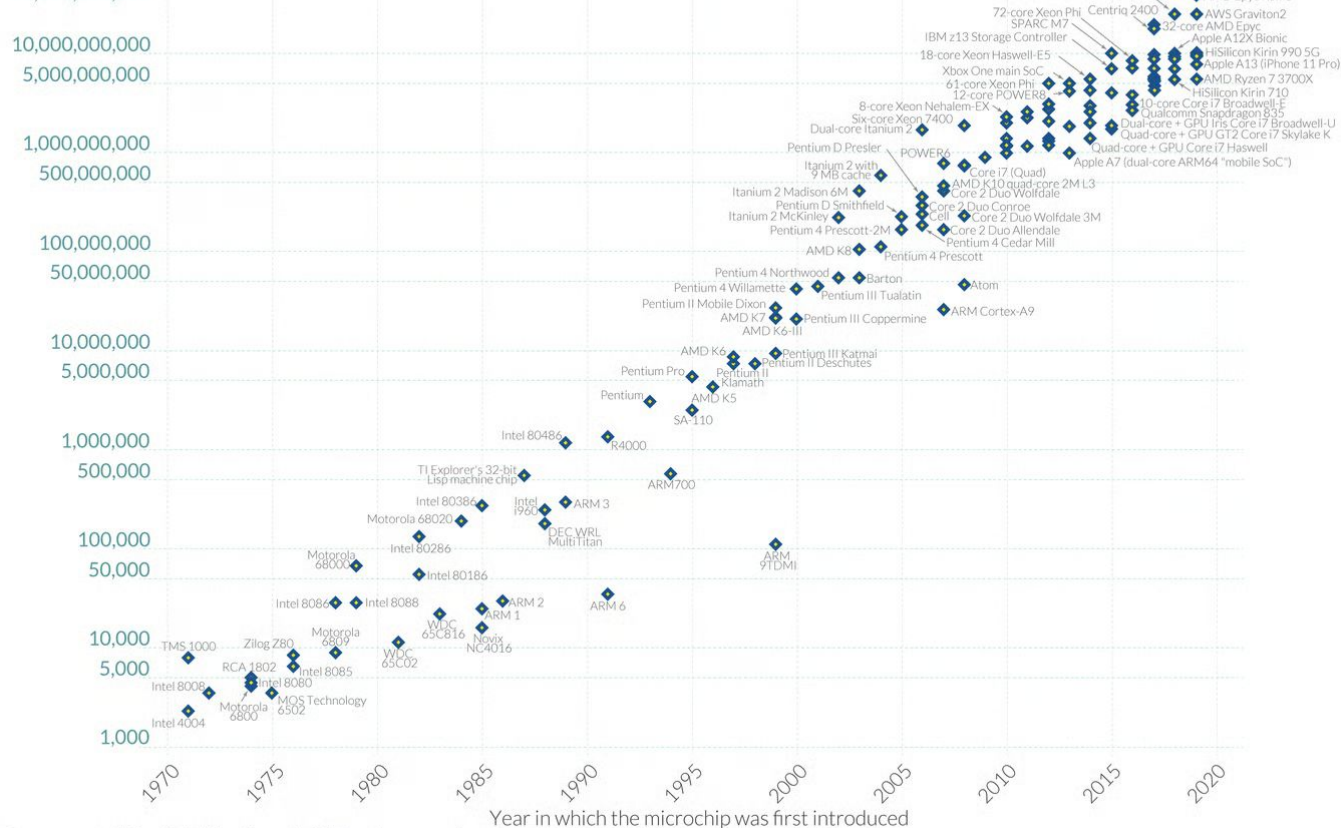
Moore's Law: The number of transistors on microchips has doubled every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Our World
in Data

Transistor count

50,000,000,000



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

OurWorldinData.org – Research and data to make progress against the world's largest problems.

Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.

Learning Goals



Learning Goals

After this lecture, you should be able to:

- Describe the **classification** steps.
- Explain the concept of a **rooted tree** and **decision tree**.
- Describe what the **general decisions** are in building a decision tree.

Classification

Classification

The idea behind classification is that we want to use **patterns** and/or **correlations** to make decisions

Classification happens all the time in real life

- The doctor uses your symptoms and other measurements like weight/blood pressure/etc. to help make a diagnosis
- Google uses classification to determine what an image is

Classification is a **general class** of algorithms.

Classifiers

Classifiers are **algorithms** that **perform classification**

They are specific

- e.g. - we don't give loans to anyone an income of less than \$50,000 per year

The algorithm you come up with is no different than the other algorithms you've come up with so far

You still need to state the steps you need to take to come up with the solution

Classifiers

Classifier

- A **classifier** is an **algorithm** that maps the input data to a specific category
 - Classifiers are derived from patterns or correlations from data.

Classifier: Training vs Test Data

- The data that classifiers learn the patterns has the “answer”
 - This data is called **training data**.
- Some of the training data is held back to check and see if the classifier works.
 - This is called **test data**.

Classifiers + Data

Classifiers then apply these patterns to new data where we don't know the answer

- **Input:** Digital image
- **Output:** Cat/not a cat
- **Training data:**
Labeled images of cats and
images that are not cats



Clicker Questions (using Agora)



Clicker Question



Q: What is an algorithm?

- A. A series of steps written to solve a specific problem with specific inputs
- B. A series of steps written to solve a specific problem with non-specific inputs
- C. A series of steps written to solve a non-specific problem with specific inputs
- D. A series of steps written to solve a non-specific problem with non-specific inputs

Classification

Task: **Loans**

An ***algorithm*** describes a sequence of steps that is:

1. Unambiguous

- No “assumptions” are required to execute the algorithm
- The algorithm uses precise instructions

2. Executable

- The algorithm can be carried out in practice

3. Terminating

- The algorithm will eventually come to an end, or halt

The idea behind classification is that we want to use **patterns** and/or **correlations** to make decisions

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Classification Task - Cancer Treatment

Input: Genome sequence from cancerous biopsy tissue

- Age, gender, race, weight, family history, overall health, etc.

Output: Which cancer treatment is likely to work best

Training data:

Labeled genome sequences and which treatments were successful from both cancerous tissue



Classification Task - Loan Applications

Input: Individual's loan application

- Address, age, gender, credit rating, etc.

Output: Accept or reject loan application.

Training data:

List of loan apps, decisions made, and for those who were approved, whether they repaid the loan or not

Building a Classifier: Loans



Class Activity (on PrairieLearn)

Building a Classifier: Loans

Here is your past (training) data
on some loans

Applicant	Income	Gave loan?
#1	\$50,005	Yes
#2	\$25,004	No
#3	\$75,005	Yes
#4	\$95,005	Yes
#5	\$45,007	No

Task: Create an algorithm to decide what your classifier does: i.e., when will you give a loan, and when will you not give a loan?

Participation Question



Q: Would your classifier give a loan if an applicant's income was \$78,000?

Yes; explain

No; explain

-

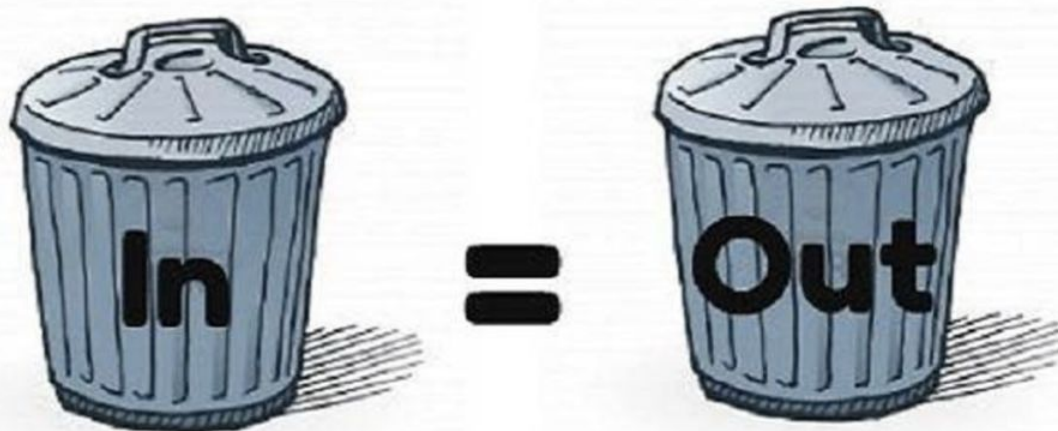
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Applicant	Income	Gave loan?
#1	\$50,005	Yes
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#3	\$75,005	Yes
#4	\$95,005	Yes
#5	\$45,007	No



Real World Examples

Training data influences the classifier



Garbage in, Garbage out (GIGO)



Insight - Amazon scraps secret AI recruiting tool that showed bias against women

By Jeffrey Dastin

October 10, 2018 5:50 PM PDT · Updated 6 years ago



SAN FRANCISCO (Reuters) - Amazon.com Inc's machine-learning specialists uncovered a big problem: their new recruiting engine did not like women.

The team had been building computer programs since 2014 to review job applicants' resumes with the aim of mechanizing the search for top talent, five people familiar with the effort told Reuters.



How We Analyzed the COMPAS Recidivism Algorithm

by Jeff Larson, Surya Mattu, Lauren Kirchner and Julia Angwin

May 23, 2016

[← Read the story.](#)

Across the nation, judges, probation and parole officers are increasingly using algorithms to assess a criminal defendant's likelihood of becoming a recidivist – a term used to describe criminals who re-offend. There are dozens of these risk assessment algorithms in use. Many states have built their own assessments, and several academics have written tools. There are also two leading nationwide tools offered by commercial vendors.

We set out to assess one of the commercial tools made by Northpointe, Inc. to discover the underlying accuracy of their recidivism algorithm and to test whether the algorithm was biased against certain groups.

Our analysis of Northpointe's tool, called COMPAS (which stands for Correctional Offender Management Profiling for Alternative Sanctions), found that black defendants were far more likely than white defendants to be incorrectly judged to be at a higher risk of recidivism, while white defendants were more likely than black defendants to be incorrectly flagged as low risk.

ARTICLE AI

In 2016, Microsoft's Racist Chatbot Revealed the Dangers of Online Conversation > The bot learned language from people on Twitter—but it also learned values

BY OSCAR SCHWARTZ | PUBLISHED 25 NOV 2019 | UPDATED 04 JAN 2024 | 5 MIN READ |





Harvard Griffin GSAS Science Policy Group

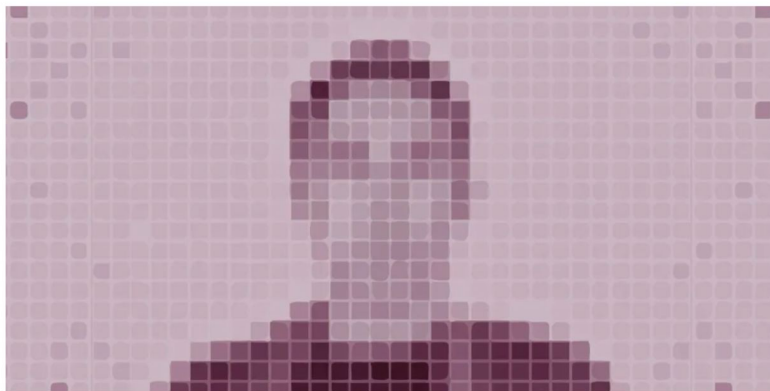
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Racial Discrimination in Face Recognition Technology

October 24, 2020



By Alex Najibi

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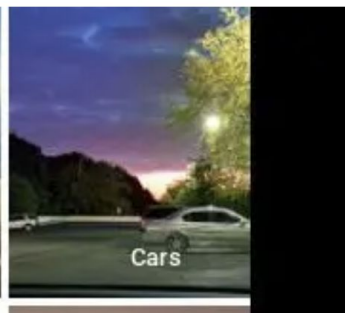
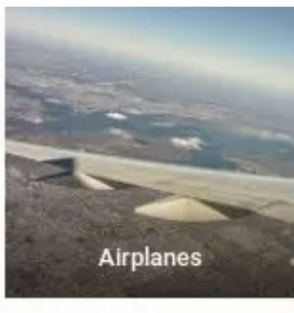
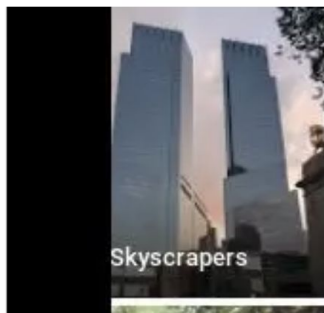
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Google apologises for Photos app's racist blunder

🕒 1 July 2015





Steps to do Classification



Step 1: Start with the data you have

Applicant	Annual Income	Loan Approved?
#1	26 000	No
#2	60 000	Yes
#3	50 000	Yes
#4	47 000	No
#5	12 000	No
#6	108 000	Yes



Step 2: Split data into training and test sets

We chose a 50/50 split for our demo but you could do other splits like 60/40, 70/30. For large datasets, 80/20 split is used

Applicant	Annual Income	Loan Approved?
#1	26 000	No
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} **Training Data**

Applicant	Annual Income	Loan Approved?
#4	47 000	No
#5	12 000	No
#6	108 000	Yes

} **Test Data**



Step 3: Build classifier

(i.e., Find pattern in training set)

Given your training data, can you find a pattern that can tell you when to approve a loan?

Earlier, we decided an annual income of ~\$50,000 seemed like a good cut off point. **That was a classifier!**

Applicant	Annual Income	Loan Approved?
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#2	60 000	Yes
#3	50 000	Yes

} Training Data



Step 4: Use classifier on test data

Applicant	Annual Income	Loan Approved?
#4	47 000	?
#5	12 000	?
#6	108 000	?

} **Test Data**

After you come up with a classifier that seems to do okay with your training data, you use it on your test data to see what kinds of decisions it makes.



Step 5: Calculate Accuracy

Applicant	Annual Income	Loan Approved?	Classifier said to...
#4	47 000	No	No
#5	12 000	No	No
#6	108 000	Yes	Yes

} **Test Data**

If the results of your classifier match up with the decisions you've made in your test data, it's looking good.

You can start trying to use it on data that you haven't made any decisions on yet.





Seems Straightforward

- What happens when we have more than one attribute?
- In the example before, we only had to consider annual income
- But what would happen if we had multiple attributes, like 5 or 10 or 100?
- **How do we decide which attribute to use?**



Seems Straightforward

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Decision Trees!



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Decision Trees!

(next class!)

Wrap up